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1. Define Machine Learning (ML). List types of ML.

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make decisions without being explicitly programmed. The system improves its performance automatically through experience. The main types of machine learning are supervised learning, unsupervised learning, and reinforcement learning.

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2. Explain exploration and exploitation in reinforcement learning and its importance.

Exploration refers to trying new actions to discover better rewards, while exploitation means choosing actions that are already known to give maximum reward. Exploration helps the agent gain knowledge about the environment, whereas exploitation helps maximize performance. A balance between both is essential for effective learning.

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3. Differentiate between machine learning and human learning.

Machine learning depends on algorithms and large datasets to learn patterns, whereas human learning depends on experience, understanding, and reasoning. Humans can learn from very few examples, but machines usually need large amounts of data. Human learning is flexible and adaptive, while machine learning is limited to the provided data and model.

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4. Identify types of machine learning used for given problems.

Intelligent robots mainly use reinforcement learning because they learn through interaction and feedback. Stock market prediction uses supervised learning as historical labeled data is available. Customer segmentation uses unsupervised learning since customers are grouped without predefined labels.

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5. Give an example of application for each type of machine learning.

Supervised learning is used in spam email detection where data is labeled.
Unsupervised learning is used in customer clustering to find hidden patterns.
Reinforcement learning is used in game playing systems and robot navigation.

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6. How to create DataFrame in Pandas.

A DataFrame is a two-dimensional data structure in Pandas used to store data in rows and columns. It is created using lists, dictionaries, or NumPy arrays and is widely used for data analysis and manipulation.

Python Code:

```python
Import pandas as pd

Data = {
  “Name”: [“A”, “B”, “C”],
  “Marks”: [70, 85, 90]
}

Df = pd.DataFrame(data)
Print(df)
```

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7. How to create a Series from list, NumPy array, and dictionary.

A Pandas Series is a one-dimensional labeled array. When created from a list or NumPy array, default indexing is applied. When created from a dictionary, keys act as index labels.

Python Code:

```python
Import pandas as pd

Import numpy as np

S1 = pd.Series([10, 20, 30])

S2 = pd.Series(np.array([40, 50, 60]))

S3 = pd.Series({“A”: 1, “B”: 2, “C”: 3})

Print(s1)

Print(s2)

Print(s3)
```

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8. Write a Pandas program to find and drop missing values.

Missing values reduce data quality and affect model performance. Pandas provides functions to detect and remove missing values so the dataset becomes clean for analysis.

Python Code:

```python
Import pandas as pd

Df = pd.DataFrame({
```

```
    “A”: [1, 2, None],
    “B”: [4, None, 6]
})

Print(df)
Df_clean = df.dropna()
Print(df_clean)
```

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9. How to use title, grid, labels, and show chart in Matplotlib.

Matplotlib allows adding titles, axis labels, and grid lines to improve graph readability. The show function is used to display the final chart.

Python Code:

```python
Import matplotlib.pyplot as plt

X = [1, 2, 3, 4]
Y = [10, 20, 25, 30]

Plt.plot(x, y)
Plt.title(“Sample Graph”)
Plt.xlabel(“X Axis”)
Plt.ylabel(“Y Axis”)
Plt.grid(True)
Plt.show()
```

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**10. Compute mean, standard deviation, and variance along second axis.**

NumPy provides statistical functions that compute values along a specific axis. Using axis=1 calculates row-wise results for a two-dimensional array.

**Python Code:**

```python
Import numpy as np

Arr = np.array([[1, 2, 3],
        [4, 5, 6]])

Print(np.mean(arr, axis=1))
Print(np.std(arr, axis=1))
Print(np.var(arr, axis=1))
```

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11. Write a Python program to implement any four maths functions in NumPy.

NumPy provides efficient mathematical functions that operate on arrays and are widely used in scientific and machine learning computations.

Python Code:

```python
```
```

```
Import numpy as np

A = np.array([1, 4, 9, 16])

Print(np.sqrt(a))
Print(np.power(a, 2))
Print(np.sin(a))
Print(np.log(a))
```

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12. Explain Pandas Series with example.

A Pandas Series is a one-dimensional data structure that stores values along with an index. It is similar to a column in a table and can hold numerical or textual data.

```
import pandas as pd

data = {
   "A": [1, 2, None],
   "B": [4, None, 6]
}

df = pd.DataFrame(data)
print(df)

df_clean = df.dropna()
print(df_clean)
```

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13. Explain pie chart, bar chart, and histogram in Matplotlib.

A pie chart represents data as proportions of a whole. A bar chart compares values across categories using bars. A histogram shows the frequency distribution of numerical data using bins.

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14. Explain steps involved in preparing the model activity in machine learning.

Model preparation includes selecting an algorithm, splitting data into training and testing sets, training the model, tuning parameters, and evaluating performance using test data.

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15. Explain types of data in Machine Learning.

Data in machine learning can be qualitative or quantitative. Qualitative data represents categories or labels, while quantitative data represents numerical values used for calculations.

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16. Calculate accuracy and error rate for cricket dataset.

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17. Define Dimensionality Reduction.

Dimensionality reduction is the process of reducing the number of input features while preserving important information. It improves model efficiency and reduces overfitting.

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18. Define Feature Subset Selection.

Feature subset selection is the process of selecting important features and removing irrelevant or redundant ones to improve model performance and reduce complexity.

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19. Explain Cross Validation.

Cross validation evaluates model performance by dividing data into multiple subsets and training the model multiple times. It provides a more reliable accuracy estimate.

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20. Explain Holdout Method.

The holdout method splits the dataset into training and testing parts. The model is trained on one part and tested on the other. It is simple but less reliable for small datasets.

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21. Describe K-Fold Cross Validation.

K-Fold cross validation divides the dataset into K equal parts. The model is trained K times, each time using one fold for testing and the remaining folds for training. The final result is the average of all runs.